

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 1 Practical No: 1

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Subject Name: Deep Learning

Problem Statement: Install and configure TensorFlow/Keras in Google Colab. Perform data preprocessing, normalization, train-test splitting, and data visualization on a sample dataset.

Algorithm:

1. Start.
2. Import TensorFlow, NumPy, and Matplotlib.
3. Load the Fashion MNIST dataset.
4. Preprocess and normalize the data.
5. Visualize the dataset.
6. Build a Sequential neural network.
7. Compile the model.
8. Train the model using epochs.
9. Evaluate the model on the test dataset.
10. Display accuracy and graphs.
11. Stop.

Code:

```
[1] # 1. Importing Libraries
✓ 9s import tensorflow as tf
import matplotlib.pyplot as plt
import numpy as np

[2] # 2. Get data and split into train and test datasets
✓ 1s (train_images, train_labels), (test_images, test_labels) = tf.keras.datasets.fashion_mnist.load_data()

> Show hidden output

[3] # 3. Display Dataset Information
✓ 0s
print("Training Images Shape :", train_images.shape)
print("Training Labels Shape :", train_labels.shape)

print("Testing Images Shape :", test_images.shape)
print("Testing Labels Shape :", test_labels.shape)

print("\nData Type :", train_images.dtype)

> Show hidden output

[4] # 4. Scale down values of pixels from 0-255 to 0-1
✓ 0s
train_images = train_images / 255.0
test_images = test_images / 255.0

[12] # 5. Visualizing data
class_names = ['T-shirt/Top', 'Trouser', 'Pullover', 'Dress', 'Coat', 'Sandal', 'Shirt', 'Sneaker', 'Bag', 'Ankle Boot']
plt.imshow(train_images[9], cmap='gray')
plt.title(class_names[train_labels[9]])
plt.axis('off')
plt.show()

> Show hidden output

[15] # 6. Training model
# Define the neural network model
model = tf.keras.models.Sequential([
    tf.keras.layers.Flatten(input_shape=(28, 28)),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(10, activation='softmax')
])
# Compile the model
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
# Train the model
history = model.fit(train_images, train_labels, epochs=3)
# Evaluate on test dataset
test_loss, test_accuracy = model.evaluate(test_images, test_labels)
```

```
print("Test Loss:", test_loss)
print("Test Accuracy:", test_accuracy)
plt.plot(history.history['accuracy'], marker='o')
plt.title("Training Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.grid(True)
plt.show()
```

> Show hidden output

```
[7] plt.figure(figsize=(8,8))

for i in range(9):
    plt.subplot(3,3,i+1)
    plt.imshow(train_images[i], cmap='gray')
    plt.title(class_names[train_labels[i]])
    plt.axis('off')

plt.tight_layout()
plt.show()
```

> ... Show hidden output

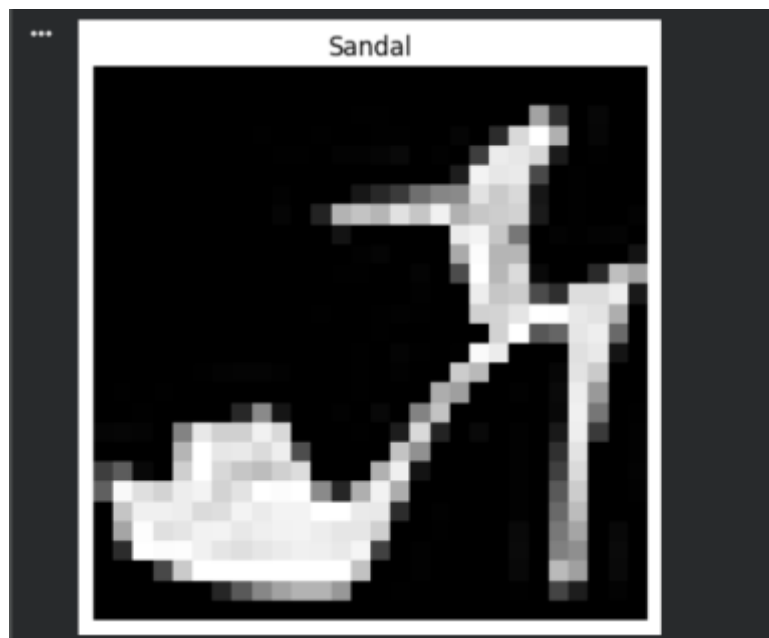
Output:

3. Display Dataset Information

```
... Training Images Shape : (60000, 28, 28)
Training Labels Shape : (60000,)
Testing Images Shape : (10000, 28, 28)
Testing Labels Shape : (10000,)

Data Type : uint8
```

5. Visualizing data



6. Training model



7. Display the classes

